

Treatment of Glomerulonephritis

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KEY POINTS

- The incidence and prevalence of glomerulonephritis (GN) is likely underestimated.
- There is a growing body of literature guiding treatment of GN based upon randomized-controlled trials.
- Decisions regarding therapy should be tailored to the individual patient, with thoughtful balance of the risk of disease progression and the risk of treatment toxicity in the individual patient.
- The development of novel therapies and new protocols to treat GN underscore the importance of mitigation of medication toxicity.
- Age and cumulative dose modulate risk of gonadal toxicity from alkylating agents.
- Mycophenolic acid has recognized teratogenicity.
- Treatment with Eculizumab is associated with susceptibility to infections with encapsulated bacteria including *Neisseria meningitidis* meningitis. Prior vaccination and prophylactic antibiotics are recommended.

INTRODUCTION

The societal burden of glomerulonephritis (GN) on both the individual and the healthcare system is substantial. Although renal replacement therapy is an essential life-sustaining therapy for patients with kidney failure, advances in the therapeutic options available for treatment of GN offer great potential to avoid kidney failure in patients afflicted with these diseases. The field of GN treatment is also progressing quickly toward individualization of therapy—through development of more targeted and less toxic therapies and

identification of patients who have the greatest potential for benefit from these medications. In this chapter we discuss strategies to individualize care of patients with GN, and opportunities to mitigate risk of currently available treatment options.

THE GLOBAL IMPACT AND CHALLENGES

Estimating the incidence of GN and its impact on patient morbidity and mortality is challenged by several barriers. First, the incidence of GN is grossly underappreciated and

Table 33.1. Considerations for Individualization of Therapy and Avoidance of Drug Toxicity

	Considerations	Example
Disease	Disease-specific treatment targets	A partial remission of proteinuria may portend favorable prognosis and minimize drug toxicity
	Disease-related risks with no treatment	Risk of thrombosis with untreated membranous nephropathy, life-threatening risks of vasculitis
Patient	Modifiable patient factors	Obesity, smoking
	Nonmodifiable patient factors	Age, comorbid conditions
	Exposures	Endemic and latent infections
Drug	Anticipation of drug-specific toxicities	Steroid avoidance in prediabetes, infection prophylaxis

the costs underestimated because of the lack of national or international registries of GN at the level of renal pathology. Although large prospective study cohorts are now established to study the natural history and pathogenesis of these diseases (reviewed elsewhere¹), the incidence of biopsy-proven GN cannot be calculated from these studies as the geographic region from which patients are enrolled is not as carefully delineated.

The extent of these diseases may, therefore, be better approximated by imputing incidence rates from country-wide renal pathology registries where there is prescribed reporting. Such a registry from Finland, for example, reported an incidence of GN that ranged from 8.7 to 25.4 per 100,000 population in the central hospitals and the university center, respectively, which was remarkably higher than the rates reported by the European Biopsy Registry (between 1 and 6.9).² The full disease impact is also substantially discounted if assessed solely from figures derived from end-stage renal disease (ESRD) registries as patients presenting with advanced chronic kidney disease (CKD) or ESRD are often not biopsied. Most recently, a study of a data set consisting of a public Medicare and two insurer databases capturing over 19 million clients also suggested that the incidence of GN may be far higher than previously estimated.³ Based upon health claim data (International Classification of Diseases, 9th Revision [ICD-9] codes), the incidence of GN was estimated to be as high as 134 per 100,000 patient-years (95% confidence interval [CI] 132–136). Rates were lower in the privately insured but remained much higher than previously referenced registry estimates.

Understanding the incidence of GN is important. As therapy becomes more tailored, the cost of medications becomes critical from a resource perspective. Health care payers cannot estimate cost–benefit without an appreciation of the incidence and course of GN. With the emergence of more costly but targeted medications, consideration of savings from avoidance of both kidney failure and complications of therapy becomes increasingly relevant. Similarly, it is incumbent upon the nephrology community to conduct rigorously designed and high-quality therapeutic trials to expedite development and quantitate the efficacy of novel therapies.⁴

PERSONALIZING THERAPY: BALANCING RISKS AND BENEFITS

GN is associated with significant morbidity and mortality that must be considered when making informed decisions

about therapy. A recent retrospective analysis of an observational cohort derived from an insurer-based (Kaiser Permanente) integrated healthcare system highlights these risks.⁵ The cohort consisted of over 2300 patients with biopsy-proven GN. At a mean follow-up of 4.5 years, 21% developed end-stage kidney failure, and 8% died before reaching this endpoint. This emphasizes the rationale for more aggressive intervention strategies on a population level. The development of the Kidney Disease Improving Global Outcomes (KDIGO) guidelines represents an important advance in harmonizing and standardizing treatment for GN and providing the evidence-based context of recommendations. However, one of the challenges of adopting clinical guidelines is individualizing treatment strategy to the patient.

A focus of our practice is developing patient-specific risk evaluation so that treatment benefits and risks can be considered at the individual patient level. This can be adopted formally through checklist-style documentation or more informally during the consultation process. Consideration of personalized treatment is critical to balance the potential risks and benefits of immunotherapy, so that both clinicians and patients can enter a treatment plan with an informed and collaborative decision making process.

Broadly, there are three primary areas for consideration when evaluating a treatment plan for an individual patient (Table 33.1). These include consideration of the specific disease-associated risks and treatment goals, the patient characteristics, and the specific medication toxicities. As one evaluates a patient, each of these categories is considered and pros and cons of various strategies are measured and reviewed with the patient—an active participant in his or her care.

PERSONALIZING TREATMENT: DISEASE-SPECIFIC RISKS AND GOALS

Treatment cannot be personalized without understanding the specific nature of the disease and risk of disease progression. This includes the disease natural history and the therapeutic goals. To balance the risks of therapy, consideration of the individual patient's likelihood of progression and the disease natural history (including patient morbidity) should be reviewed.

Risk of disease progression is described largely in natural history studies that have yielded important information that has influenced patient management and altered how we assess treatment benefit.^{6–9} These efforts have allowed a more accurate estimation of the impact of modifiable predictors

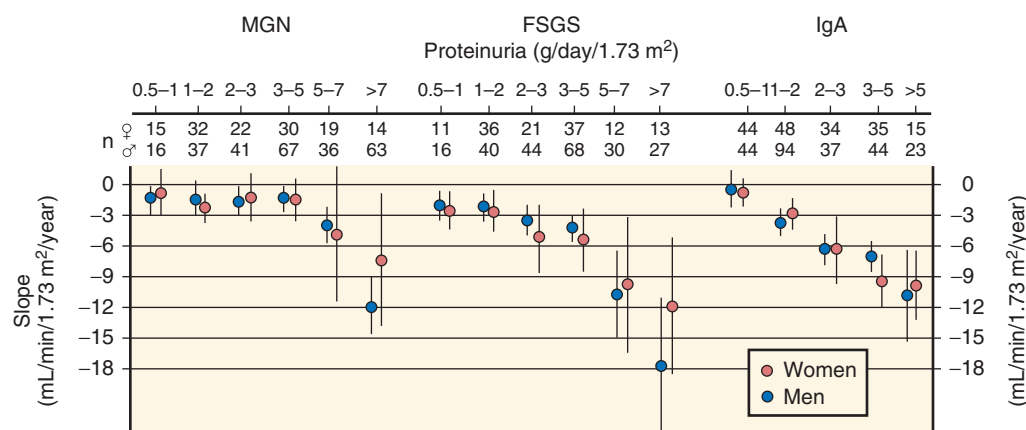


Fig. 33.1 Differential impact of proteinuria according to sex and histologic subtype on the rate of renal function decline in membranous glomerulonephritis (MGN), focal and segmental glomerulosclerosis (FSGS), and immunoglobulin A (IgA) nephropathy. This figure illustrates the annual rate of renal function decline (y-axis) according to categories of time-averaged proteinuria (x-axis). This is described by disease subtype and gender. (From McCarthy ET, Sharma M, Savin VJ. Circulating permeability factors in idiopathic nephrotic syndrome and focal segmental glomerulosclerosis. *Clinical Journal of the American Society of Nephrology*: CJASN. 2010;5(11):2115-2121.)

Table 33.2. Definition of Remission and Outcome by Histologic Diagnosis in Primary Glomerulonephritis

	Membranous Glomerulonephritis	Focal and Segmental Glomerulosclerosis	Immunoglobulin A Nephropathy
Definition of partial remission	Reduction to <3.5 g/day and 50% decrease from peak proteinuria	Reduction to <3.5 g/day and 50% decrease from peak proteinuria	Reduction to <1 g/day proteinuria
Rate of renal function decline	CR: -1 mL/min/year PR: -2 mL/min/year NR: -10 mL/min/year	CR: -0.1 mL/min/year PR: -5.6 mL/min/year NR: -10 mL/min/year	CR: -1.6 mL/min/year PR: -2 mL/min/year NR: -9 mL/min/year

CR, Complete remission; NR, no remission; PR, partial remission.

on clinical outcome, thus allowing physicians to provide general and disease-specific recommendations. For example, hypertension is a risk factor for progression across all forms of primary and secondary GN, and targets may be relatively generalizable across kidney disease categories. However, the relationship between quantity of proteinuria and outcome in GN varies across diagnoses.

Traditionally, clinicians have estimated the risk of progression of GN based upon whether a patient has proteinuria in the nephrotic range or not. Comprehensive reviews of large cohorts of patients with biopsy-proven GN have highlighted that proteinuria has different implications dependent upon the underlying diagnosis. This is illustrated in Fig. 33.1. As can be seen in this figure, the rate of renal function decline according to category of sustained proteinuria does not largely differ according to patient sex, but varies greatly according to histologic diagnosis. A finding of 2 g/day of proteinuria in a patient with immunoglobulin A nephropathy (IgAN) portends a far worse prognosis than in a patient with membranous GN (MGN). Nephrotic versus nonnephrotic categorization of proteinuria is therefore too simplistic for estimating disease risk.

Natural history studies have also helped to define treatment goals and the value of proteinuria reduction in GN (Table 33.2). The quantitative value of proteinuria reduction is a crucial element in the decision making of nephrologists in

terms of balancing the risks and benefits of treatment and is emerging as a surrogate endpoint for renal survival. The capacity to translate proteinuria reduction into a disease-specific estimate of improvement in long-term outcome provides an important element of the benefit-risk equation not only in terms of whether to initiate treatment, but also more commonly today to provide help in the decision about prolonging treatment or retreating a patient to maintain or reestablish a partial remission.

MEMBRANOUS GLOMERULONEPHRITIS

Natural history studies in MGN have demonstrated the value of complete remission of proteinuria. A prospective study of 348 nephrotic patients with MGN assessed the relative benefit of a complete, or partial, remission of proteinuria on both renal survival and rate of progression.⁸ Partial remission was defined as both a reduction to less than 3.5 g/day and a 50% decrease from peak proteinuria. Over a median follow-up of 60 months, 30% had a complete remission, 40% had a partial remission, and the remaining 30% had no remission. At 10 years, renal survival in those with a complete remission was 100% with minimal loss of function over time (rate of loss of creatinine clearance -1 mL/min/year). Those achieving a partial remission had a 90% renal survival at 10 years, and a more rapid rate of progression compared with those

with a complete remission, although still limited to a loss of -2 mL/min/year of creatinine clearance. In comparison, those with no remission had a renal survival of only 50% at 10 years and a significant fivefold rate of renal function loss compared with the partial remission group (-10 mL/min/year). Survival from renal failure for partial remission was significantly improved (hazard ratio for ESKD vs. the reference group of no remission of 0.08, 95% CI 0.03–0.19; $P < .001$). This study, however, did not identify the intervention required for achievement of partial remission of proteinuria; partial remission of proteinuria is also associated with favorable outcome when achieved spontaneously.¹⁰

A more recent analysis of the same cohort further reinforces the value of proteinuria reduction. Indeed, the quantity of time spent in either complete or partial remission imparts better preservation of renal function and a lower rate of kidney failure.¹¹ There is increasing support that partial remission of proteinuria is a surrogate marker of improved “hard” renal outcomes such as survival free from kidney failure.¹²

FOCAL AND SEGMENTAL GLOMERULOSCLEROSIS

The same estimates of benefit of proteinuria reduction on long-term outcome are observed in patients with focal and segmental glomerulosclerosis (FSGS). It has been appreciated for some time that complete remission of proteinuria is the best predictor of a favorable renal survival.^{13,14} Factors that have been previously associated with a poor outcome in FSGS have included the severity of initial proteinuria, renal function at presentation, and the extent of tubulointerstitial disease on histologic examinations.^{10,15} Proteinuria reduction is an independent determinant of outcome.¹⁶

The impact of partial proteinuria reduction on disease course was addressed in a long-term cohort study of 281 nephrotic patients with biopsy-proven primary FSGS followed over an average of 65 months.⁹ A partial remission is defined by both a reduction of proteinuria to the subnephrotic range (<3.5 g/day) and a 50% reduction from peak proteinuria. During the observation period 55 patients had a complete remission, 117 patients achieved a partial remission, and 109 had no remission of proteinuria. Partial remission was independently predictive of both renal survival and rate of decline in renal function, and was associated with more favorable outcome. Partial remission was associated with improved renal survival with a time-adjusted hazard ratio of 0.48 (95% CI 0.24 to 0.96; $P = .04$). Ten-year renal survival was 75% in the partial remission group compared with 35% in those with no remission. Similar to MGN, this information on the quantitative benefits of partial remission is important in the assessment of the FSGS patient because it provides equipoise for the risks of treatment.

IGA NEPHROPATHY

Proteinuria, at far lower levels compared with MGN and FSGS, is tightly correlated with renal outcome in patients with IgAN.¹⁷ Treatment and observational studies support the importance of proteinuria reduction in mitigating risk of progressive disease in IgAN. Donadio et al.¹⁸ demonstrated an association between proteinuria reduction and both improved renal survival and prolonged time to doubling of serum creatinine

in patients enrolled in a trial of fish oil. A cohort of more than 500 patients with primary IgAN nephropathy followed for an average of 78 months evaluated the clinical relevance of achieving a partial remission of proteinuria to <1 g/day. In the almost 200 patients that achieved and sustained partial remission of proteinuria (either spontaneously or through treatment), the mean rate of decline in renal function was only 10% of those who did not. Furthermore, regardless of the level of presenting proteinuria, those that attained a partial remission had the same long-term prognosis and slow rate of disease progression as those patients whose peak proteinuria never exceeded 1 g/d. Although there were other modifiable factors identified in the multivariate analysis associated with kidney function decline (time-averaged mean arterial pressure and exposure to renin–angiotensin system blockade), the level of sustained proteinuria was the dominant modifiable risk. The differential in progression rate and renal failure risk was dramatic and understanding the impact of even a small but sustained improvement in proteinuria is extremely valuable information for the practicing physician. Occasionally IgAN presents with the nephrotic syndrome and preserved renal function. In this subset of patients, partial or complete remission of the nephrotic syndrome is also associated with a favorable outcome.¹⁹

Proteinuria has strong links to cardiovascular mortality in patients without GN.²⁰ A longitudinal study of nearly 1400 patients with IgAN followed in Korea suggested that this disease—particularly when associated with proteinuria greater than 1 g/day—may be associated with a higher standardized mortality rate.²¹ The potential independent benefit of proteinuria reduction on mortality certainly merits further study.

The Supportive Versus Immunosuppressive Therapy for the Treatment of Progressive IgAN (STOP-IgAN) study suggested a disconnect between proteinuria reduction and renal survival.²² Although more patients achieved remission of proteinuria after having received corticosteroids, this did not translate to improved long-term outcome. One possible explanation for this disconnect may be that overall the cohort had relatively low-grade proteinuria, and a very low rate of renal function decline. Therefore it would be difficult to demonstrate differences in “hard” renal endpoints. In addition, the treatment lasted only 6 months. Proteinuria reduction may not always be maintained and 6 months may be insufficient time to accrue the benefit of proteinuria reduction over the long term in this population.

REMISSION IN LUPUS NEPHRITIS AND VASCULITIS

Estimation of the value of proteinuria remission on outcome is more complex in GN secondary to vasculitis and lupus. In secondary GN the potential morbidity associated with nonrenal manifestations may be the primary determinant for estimating patient risk and guiding decisions regarding immunotherapy. Cohorts differ greatly in terms of renal disease severity, serologic risk factors, and treatment regimens, making it challenging to distinguish the independent value of proteinuria reduction beyond these important factors. Long-term observational studies confirm the association of sustained proteinuria with long-term outcome.^{23–26} However, given the relapsing and systemic nature of these diseases, even complete remission of proteinuria does not guarantee

freedom from morbidity and mortality resulting from these multisystem diseases, including developing CKD.²⁷

TIMING OF RESPONSE

Expectations regarding the timing of treatment response should be well understood by patients and clinicians at the start of therapy. This helps avoid both excessive exposure to toxicities of therapy and premature escalation or interruption of a treatment protocol. For example, in the case of MGN, the earliest signs of response may only be evident at 3 months of immunotherapy. This is, in part, because of the extensive filtration barrier injury that must be remodeled to restore function. Improvement may continue for months following initiation of therapy. In the case series describing use of rituximab, the median time to remission endpoint from start of therapy was 7 months with a maximum duration extending over a year (interquartile range 3–12).²⁸ At 3 months, one should expect to see a trend toward improvement. However, evaluating the full response to a treatment can take 12 months.

Similarly, the time to normalization of urine sediment (i.e., resolution of hematuria) is often delayed in secondary GN. At 6 months of evaluation, less than half of patients with severe lupus nephritis enrolled in therapeutic trials will have reached a complete remission according to definitions that include renal sediment findings.²⁹ A recent review of long-term outcomes of patients with severe lupus nephritis treated in a prior randomized controlled study indicated that halving of proteinuria at 6 months was an important predictor of both long-term renal and patient survival.³⁰ Similarly, in a renal-focused trial of rituximab in vasculitis, the mean time to complete remission was 182 days (standard deviation 43 days) in the rituximab group and 202 days (standard deviation 66 days) in the cyclophosphamide/azathioprine group.³¹ These findings emphasize that provided there is evidence of improvement and treatment is being well tolerated, patience is warranted before assessing for full medication response.

Disease-specific treatment goals are critical to planning individualized therapy and minimizing toxicity. However, untreated disease can also be associated with substantial morbidity and mortality. For example, a 6-month observation is commonly the first step in management of membranous nephropathy. Indeed, with prolonged observation spontaneous remission can occur beyond a year, and can occur even in patients with high-grade proteinuria.¹⁰ However, prolonged periods of profound hypoalbuminemia will place patients at added risk of not only infections, and thrombotic events despite sparing exposure to immunosuppression, but also of progression of parenchymal damage during this waiting period. Therefore these competing risks must be weighed in each patient. Strategies to mitigate risks of postponing treatment (e.g., prophylactic anticoagulation in MGN) should also be factored into plans. Similarly, patients with advanced kidney injury with vasculitis may not have salvageable renal function and certainly toxicity of immunotherapy is higher in patients on dialysis. However, vasculitis is a systemic disease with potential for extrarenal disease-associated morbidity. Avoidance of immunosuppression may not be possible.

INDIVIDUALIZING THERAPY: MITIGATING TOXICITIES OF THERAPY

With every one of the immunosuppressive agents used in the treatment of GN, the risks of treatment versus potential benefits must be assessed on the basis of drug exposure (a composite of dose and duration), and individual patient factors including gender, age, ethnicity, country of origin, and comorbid conditions such as obesity, diabetes mellitus, and cardiac disease. Further, there is no universal availability of many of the newer drugs and biologic agents because of the paucity of controlled data and the high costs of many of these agents. As such, careful attention must be paid to potential side effects from the therapeutic choices made by practicing clinicians with side-effect profiles often dominating the choice of therapy.

CORTICOSTEROIDS

Glucocorticoids (GCs) are the most common antiinflammatory and immunosuppressive drugs used in the treatment of both primary and secondary GN. They have protean effects on immune responses mediated by T and B cells, including reversibly blocking T-cell and antigen-presenting cell–derived cytokine and cytokine-receptor expression. Their hydrophobic structure permits them to easily diffuse into cells and bind to specific cytoplasmic proteins, facilitating translocation of these proteins into the cell nucleus where they bind to a highly conserved GC receptor DNA-binding domain (the GC response element) and modulate gene transcription.³² Some of the downstream effects accounting for the antiinflammatory activity of GCs include the inhibition of synthesis of proinflammatory cytokines implicated in glomerular and tubulointerstitial injury, such as interleukin-2, -6, -8 and tumor necrosis factor.^{33,34} GCs also exert a host of nontranscriptional immunomodulatory effects on immune effector cells including alteration of leukocyte trafficking and chemotactic properties, and modulation of endothelial function, vasodilatation, and vascular permeability.³² Conditions where resolution of proteinuria can be even more rapid than expected based upon immunomodulatory function (such as minimal change disease) effects on podocyte function may be more relevant. This may include alteration in podocyte function and motility.^{35–37}

MAJOR ADVERSE EFFECTS

Infection is a potential risk common to all immunosuppressant medications. Pronounced suppression of cell-mediated immunity results from the protean effects of corticosteroids on the immune system. GC exposure poses a significant short- and long-term risk of infection, particularly in older patients. A nested case–control analysis indicated a rate of serious infection as high as 46% with 6 months of continuous use of more than 5 mg/day in patients with rheumatoid arthritis.³⁸ In addition to the potential to cause infection, GCs have widespread systemic side effects, including but not limited to, impaired glucose tolerance, cardiovascular and gastrointestinal (GI) toxicity, potentially severe musculoskeletal damage as well as a large array of cosmetic, ophthalmological, and psychiatric side effects.

GCs affect glucose metabolism by increasing hepatic gluconeogenesis and decreasing peripheral tissue insulin sensitivity. These changes in glucose homeostasis may be ameliorated by dose reduction.³⁹ However, the metabolic effects of these drugs may not be completely reversible, even when the dose is reduced to physiologic range or discontinued entirely.⁴⁰ Whereas higher doses of GCs are associated with a higher risk of hyperglycemia, additional risk factors for steroid-induced hyperglycemia include African-American and Hispanic ancestry, obesity (defined as a body mass index $>30 \text{ kg/m}^2$), older age, a family history of diabetes, and the presence of other components of the metabolic syndrome.⁴⁰

GCs have important musculoskeletal effects. Muscle injury associated with chronic steroid treatment with GC produces a pattern of proximal weakness, atrophy, and myalgia. The ideal management includes discontinuation of steroid administration, although recovery can take weeks or months. Steroid myopathy is more common when the patient has been exposed to the potent fluorinated steroids (dexamethasone, betamethasone, triamcinolone), but similar patterns of muscle injury have been described with the nonfluorinated steroid such as prednisone.⁴¹

Corticosteroids have major effects on trabecular bone structure. Up to 10% of patients on long-term therapy will have a clinically evident fracture,⁴² and risk of bone fracture is increased at doses as low as 2.5 to 7.5 mg/day,⁴³ rapidly declining toward baseline after stopping treatment.

Avascular necrosis is a clinically and pathophysiologically different type of bone injury than the loss of bone density. It is a devastating condition associated with destruction of the head of the femur or other long bones. The relationship between development of avascular necrosis and dose of prednisone is less clear.⁴⁴

Vision may be affected by cataract formation, and increased intraocular pressure. Thinning of the skin, easy bruising, development of striae, and impaired wound healing may also be potentiated by GCs. Mood lability and insomnia induced by GCs also contribute to their relatively poor patient tolerance. GI effects of GCs include induction of gastritis and GI bleeding.

Given the elevated risk of cardiovascular disease in patients with kidney disease, the added cardiovascular toxicity of GCs is also an important consideration. A large cohort study of 68,781 GC users demonstrated that high-dose steroids are independently associated with cardiovascular events after adjustment for other traditional risk factors, including hypertension, glucose intolerance, and obesity.⁴⁵

STRATEGIES FOR REDUCING TOXICITY

There are several strategies available for minimizing steroid exposure. Alternate day steroids has been described to treat nephrotic syndrome.⁴⁶ However, cumulative dose does not differ with this approach, and efficacy and minimization of side effects is not proven in adults. An alternate strategy is shortening the course and/or a more rapid taper of the prednisone, and this approach is currently being investigated in the context of vasculitis (PEXIVAS study, www.clinicaltrials.gov). More commonly, a second non-GC immunosuppressive agent is introduced for its “steroid-sparing” potential. The introduction of these agents has allowed the total exposure to corticosteroids in many of these disorders to be limited

by allowing a shorter initial total exposure to the drug. In patients at very high risk of complications, consideration may also be given to use protocols classified as “second line” to avoid toxicity. For example, a calcineurin inhibitor (CNI) may be considered as a second-line therapy for nephrotic syndrome as a result of FSGS in adults according to 2012 KDIGO guidelines.⁴⁷ However, consideration may be given to using this agent as a first-line approach in patients with significant contraindications to GCs such as preexistent significant obesity, or a history of psychosis.

Alternate strategies specifically focus on reducing or preventing the complications related to corticosteroid treatment. Such prophylactic strategies include the use of antibiotics such as trimethoprim-sulfamethoxazole to prevent *Pneumocystis jirovecii* pneumonia (previously known as pneumocystis pneumonia PCP). Retrospective studies have indicated that a corticosteroid dose equivalent to 16 mg of prednisone for a period of 8 weeks was associated with a significant risk of pneumocystis pneumonia.^{48,49} Ongoing surveillance for diabetes is required along with regular eye examinations. Antihypertensive regimens may require adjustment while on high-dose therapy, and blood pressure should be evaluated regularly. Gastric protection in the form of a proton-pump inhibitor should be prescribed.

In relationship to the risk of fracture, if daily corticosteroids (0.5–1 mg/kg) are expected to be used in excess of 8 to 12 weeks in adults, consideration should be given to adding vitamin D 1000 U/day and 1 g/day of calcium, and some populations may merit more intensive therapy including addition of bisphosphonates. The American College of Rheumatology recently updated recommendations on the prevention of osteoporosis in patients on prolonged therapy.⁵⁰ These guidelines review general universal strategies to minimize risk such as optimizing calcium and vitamin D intake, and exercise. The decision to give bisphosphonates is guided by early bone density evaluation, age, and patient-specific risks that can be assessed using validated online risk calculators. Bisphosphonates may have more potent effects to prevent reduction in bone density during corticosteroid use; however, it is important to note that bisphosphonates remain in mineralized bone for months to years, posing a theoretical risk of teratogenicity when administered to women of child-bearing potential.⁵¹

CALCINEURIN INHIBITORS

Cyclosporine and tacrolimus are CNIs that suppress the immune response by downregulating T-cell activation. They specifically block calcium-dependent T-cell receptor signaling transduction, thereby inhibiting the transcription of interleukin-2 as well as other proinflammatory cytokines, in both T cells and antigen-presenting cells.⁵² Interleukin-2 serves as the major activation factor for T cells and a key modulator of both T- and B-cell activity in numerous immunological processes.⁵³ Tacrolimus and cyclosporine have a common mechanism of action (i.e., inhibition of calcineurin phosphatase), although they bind different intracellular proteins. These intracellular proteins belong to the immunophilin family with cyclosporin binding cyclophilin and tacrolimus binding FKBP12.⁵⁴ The role of differential immunophilin binding in the mechanism of toxicity is not clear, but it

may allow for the unique side-effect profile of each of these drugs. More recently an alternative mechanism of action of these agents has been suggested relating to their capacity to stabilize the internal cytoskeletal structure of the glomerular podocyte.⁵⁵ This is an intriguing possibility and may help explain its efficacy in some of the glomerular-based diseases at lower drug levels compared with those required in solid organ transplantation.⁵⁶

MAJOR ADVERSE EFFECTS

The CNI class of agents have significant adverse effects with the most concerning being its nephrotoxicity. This is particularly relevant when prolonged therapy is being contemplated. These longer treatment courses are usually given to prevent or modify the well-recognized risk of relapse of nephrotic syndrome that does occur upon treatment withdrawal. CNI-associated nephrotoxicity can be severe, and reports indicate a significant risk of CKD if the drug is given in high doses for prolonged periods, such as occurred in early recipients of nonrenal solid organ transplants.⁵⁷ However, the cyclosporine dose and duration used in these studies are no longer considered appropriate⁵⁸ and lower doses in the glomerular-based diseases versus solid organ transplant are currently advocated.⁵⁹ In addition, modifications have occurred in drug formulation of cyclosporine, which has resulted in more consistent and predictable pharmacokinetics, allowing at least potentially even lower drug exposure regimens.⁵⁹

New onset or worsening of “hypertension” is another important and common dose-dependent adverse effect of CNI use and likely contributes to their long-term nephrotoxic potential. The reported incidence of hypertension in patients with glomerular diseases treated with CNIs varies from 10% to 30%.⁶⁰ An additional significant adverse effect is the induction of glucose intolerance and even overt diabetes.⁶¹ This seems to be specific to CNIs and is somewhat more common with tacrolimus.⁶² The transplantation literature highlights the potential contributions of CNIs to development of hyperglycemia, which is thought to be a result of both impaired insulin secretion and increased insulin resistance. The higher rate of hyperglycemia associated with tacrolimus use may reflect differential effects on pancreatic beta-cell insulin transcription and release. Even when CNIs are used as monotherapy, the risk of new-onset diabetes has been reported to be as high as 4%.⁶³

As with all immunosuppressive agents, CNIs affect immune surveillance and are associated with an increased rate of infections and malignancy. The incidence of malignancy induced by CNIs alone in the glomerular diseases is very hard to determine from the literature. Very few of the GN treatment studies have been long enough to assess CNI exposure as an independent risk factor. Cyclosporine, however, has been used in the long-term management of other autoimmune diseases including rheumatoid arthritis and psoriasis. When patients with rheumatoid arthritis treated with cyclosporine were compared with control patients (who received placebo, D-penicillamine, or chloroquine) an increased cancer risk was not seen.⁶⁴ A recent review of patients with psoriasis did find an increase in the standardized incidence ratio of cancer in patients treated with cyclosporine compared with the general population. However, when examined more closely and skin malignancies, known to be

more common in patients with psoriasis, were excluded, the incidence was not significantly higher in the CNI-treated patients.⁶⁵ As such, although not well-described, the incidence of drug-associated malignancy specifically in the context of GN therapy is considered to be relatively low.⁶⁶ The underlying risks of infection associated with untreated nephrotic syndrome and the potential for malignancy associated with membranous nephropathy further complicate the assessment of risk of these complications with CNI treatment.

Other common adverse effects of CNIs are cosmetic and include gum hypertrophy and hypertrichosis (less frequent with tacrolimus than cyclosporine). The excess hair growth can be severe and is likely to contribute to poor drug adherence, particularly in young female subjects. A cohort of approximately 200 pediatric nephrotic patients treated with cyclosporine for an average of 22 months⁶⁷ was recently reviewed; reported side effects of such prolonged therapy included hypertrichosis (52.3%), gum hyperplasia (25.4%), hypertension (18.8%), and renal impairment (9.1%). Close examination of the subgroup of patients with renal impairment in this study is revealing. In the small number ($N=18$ patients) that demonstrated renal impairment, 12 recovered completely after the cyclosporine was stopped, three experienced stable but continued renal impairment, and only three (1.5% of the total number exposed) had slow progression of their renal disease. On multivariate analysis, resistance to the cyclosporine treatment was the only factor predictive of renal impairment.

STRATEGIES FOR REDUCING TOXICITY

In contrast to transplantation, long-term low-dose therapy with cyclosporine (1.0–2 mg/kg per day) with or without low-dose steroids has been shown to be both safe and effective at maintaining remission. The lower toxicity of CNIs in patients with GN is at least in part related to the lower daily maintenance dose required and the capacity to gradually increase the dose over days or weeks to achieve a therapeutic affect versus the need for a much more rapid dose escalation following solid organ transplant.⁵⁹ Although higher doses of cyclosporine may be required for the induction phase in membranous nephropathy, the initial dose can usually be reduced during the maintenance phase.⁶⁸ In renal transplant patients, CNI dose has been safely reduced after the first year with renal function remaining stable even after 20 years of exposure to this agent.⁶⁹ Nevertheless, nephrotoxicity is a risk with this therapy and careful monitoring of drug levels, a constant awareness of drug interactions (that may either increase or decrease drug levels), and frequent monitoring of renal function are mandatory.

One of the mechanisms of the nephrotoxicity that is attributable to these agents is their renal vasoconstrictive properties. This hemodynamic effect is both dose dependent and reversible,⁷⁰ but may still result in dangerous episodes of acute kidney injury, particularly when a patient is also taking concomitant renin–angiotensin system blockade. As such, patients should be cautioned as to what to do in the event of unanticipated volume contraction secondary to volume depletion. For example, patients with significant acute GI illness may be counseled to hold one dose of medication. If they are unable to safely restart the next scheduled dose, they should present for assessment and possible intravenous

hydration. The more delayed chronic damage in the tubular interstitial compartment and the small arterioles is less well understood but may also be abrogated at least in part by a dose reduction or discontinuation of the agent. CNI toxicity may be more evident when used in patients with more advanced renal impairment and/or in those with significant tubular interstitial and/or vascular changes noted on histology. However, with careful monitoring and the slow escalation of dosage, even patients with significant renal dysfunction can be safely treated with CNIs.⁷¹

The hypertension that commonly accompanies treatment with CNIs is another adverse effect that requires attention. However, the adjustments in antihypertensive medication are usually straightforward, and its presence does not generally preclude or limit CNI usage. The vasoconstrictive effects of cyclosporine, in addition to their effects on renal potassium secretion, may limit the ability to use higher doses of inhibitors of the renin–angiotensin system to control blood pressure in patients on cyclosporine. With respect to the risk of diabetes, ongoing vigilance by the prescribing physician is required. In patients at highest risk for developing glucose intolerance including obese individuals, those with a strong family history of diabetes, older age, and metabolic syndrome,⁴⁰ strategies for preventing this adverse effect include preferential use of cyclosporine over tacrolimus and/or the use of CNI monotherapy or in combination with other steroid-sparing agents, thereby at least avoiding the additive risk of corticosteroid exposure.

ALKYLATING AGENTS

Cyclophosphamide is the most common drug in this class used in the treatment of GN. It is a cytotoxic agent that acts largely through the alkylation of purine bases. This DNA damage induces apoptosis or altered function of both B cells and T cells.^{72,73} Chlorambucil is the other drug of this class that is used, although less commonly than cyclophosphamide because of significant differences in both the short- and long-term adverse effect profile and drug tolerability.

MAJOR ADVERSE EFFECTS

Infertility in both men and women has been reported with these agents, most commonly following cyclophosphamide and is likely the most concerning long-term side effect given the often younger age of patients with glomerular disease. This effect is closely related to total exposure but is also strongly impacted by the age of the patient.⁷⁴ One early series indicated that the rate of permanent ovarian failure was 26% in those who received between 10 and 20 g cyclophosphamide, but was greater than 70% in those whose cumulative dose was more than 30 g.⁷⁵ This effect is of particular concern in women who are increasingly delaying conception to the later part of their reproductive life because of career demands. It has been estimated that women who receive a single course of cyclophosphamide therapy (10- to 20-g exposure) before the age of 25 years are at significantly less risk of permanent sterility (0%–15% risk) compared with the same exposure after the age of 30 years (30%–40% risk).^{75–77} Similar drug exposure, but an even higher risk has been estimated by Ioannidis et al. who calculated the risk of permanent ovarian

failure for a standard dose of 12 g/m² to be 90% in women when treated after the age of 32.⁷⁸ This combination of age and exposure on fertility can be expressed as an odds ratio. These authors suggest an odds ratio for permanent ovarian failure of 1.48 per 100 mg/kg of cumulative dose and 1.07 per patient-year.⁷⁹ Although more difficult to estimate, there is certainly a substantial risk of infertility in men as well. However, the age effect has not been as clearly demonstrated as in women. Studies have indicated that long-term gonadal toxicity was not evident until the cumulative exposure to cyclophosphamide was greater than 300 mg/kg, but more recent information suggests a substantial risk at a cumulative dose of less than 168 mg/kg (equivalent to 12 g for a 70-kg patient),^{80,81} and gonadal toxicity as indicated by a reduction in sperm count has been documented with exposures as low as 100 mg/kg.⁷⁴

The other major adverse effect is the risk of malignancy. It is suspected that this has been underestimated in the past at least in part because of the delay between exposure to the drug and the appearance of the cancer. This latent period may be many years. More recent data from an epidemiological study of 293 Danish patients with antineutrophil cytoplasmic antibody-associated vasculitis treated with cyclophosphamide suggested a much lower safety limit for exposure than previously indicated.⁸² The authors of that study concluded that patients who received a cumulative dose of more than 36 g of cyclophosphamide (equivalent to 2 mg/kg for 8 months in a 70-kg patient) had a substantially increased risk for the development of a malignancy compared with the normal-age and sex-controlled population. Their standardized incidence ratio of acute myelocytic leukemia was 59.0, bladder cancer was 9.5, and nonmelanoma skin cancer was 5.2 above this cumulative cyclophosphamide exposure. They also confirmed the substantial delay between exposure and malignancies. The latent period after cyclophosphamide and malignancy diagnosis averages between 6.9 and 18.5 years, and patients therefore require lifelong screening for malignancies. This exposure of 36 g is a much lower threshold for these serious complications than previously estimated and needs to be validated in an independent data set.^{83,84} In the meantime, however, the potential for toxicity at much lower exposure limit should be kept in mind when considering the more prolonged course of cyclophosphamide as a treatment option in membranous nephropathy, lupus nephritis, or vasculitis.

An additional well-recognized, short-term adverse effect of the alkylating agents is bone marrow suppression, particularly the white cell line. A recent meta-analysis reported significant leucopenia in 25% of patients with lupus nephritis who were treated with cyclophosphamide.⁸⁵ Another short-term adverse effect of cyclophosphamide is an increased susceptibility to infections. These infections can be severe and resistant to therapy and in combination with leucopenia can be overwhelming. Additional less serious but disconcerting side effects that can impact on compliance include alopecia and hemorrhagic cystitis. This long list of potentially serious complications makes monitoring, for both short- and long-term effects of these agents, a critical and necessary component of management.

Chlorambucil is an alternative alkylating agent used in the treatment of membranous nephropathy. The original regimen developed by Ponticelli et al.⁸⁶ with this agent cycled it monthly, alternating with corticosteroids over a 6-month

treatment course. The adverse effects of chlorambucil are similar to cyclophosphamide, although it has not been associated with the bladder toxicity with the associated gross hematuria. Even so, chlorambucil may be less well tolerated overall than cyclophosphamide and has the added associated risk of acute myelogenous leukemia.⁸⁷

STRATEGIES FOR REDUCING TOXICITY

Strategies to limit exposure focus on limiting duration of therapy rather than modifying the dose. The exception to this is the use of intravenous cyclophosphamide in lupus and vasculitis wherein less frequent and smaller doses of intravenous cyclophosphamide appear to be as effective as the earlier higher-dose regimens with fewer adverse events.^{73,88} A shorter-duration regimen of exposure is an established option in membranous nephropathy. The two published effective regimens vary dramatically in terms of cyclophosphamide exposure. In the original classic 6-month regimen for example, cyclophosphamide exposure is limited to 3 months⁸⁶ compared with the later published routine that employs a full year of exposure.⁸⁷ This is frequently modified to be of only 6 months' duration, but the long-term results of this reduction are still unknown.

Substitution of other agents less toxic than cyclophosphamide is another option. Mycophenolate mofetil (MPA) or azathioprine for maintenance therapy in lupus nephritis are well-established options and appear to have similar efficacy with significantly fewer adverse effects. Data from a recent randomized controlled trial in patients with diffuse proliferative lupus nephritis confirmed that MPA (3 g/day) was associated with fewer pyogenic infections than a cyclophosphamide-based regimen (relative risk 0.36).⁸⁹ Similarly, long-term therapy with azathioprine has been proven to be as efficacious as cyclophosphamide in the maintenance phase of vasculitis, with less toxicity.⁹⁰ Similar results in terms of complete and partial remissions were obtained when MPA was substituted for the year of cyclophosphamide in membranous nephropathy, but a significantly lower incidence of serious side effects was observed. This study was not a randomized controlled trial because the cyclophosphamide-treated patients were a historical control group. Although the initial response rate was equal, the relapse rate was very much higher than in the cyclophosphamide group.⁹¹ Replacement of the cyclophosphamide with CNIs is another option. This substitution strategy could be employed when initial therapy with cyclophosphamide has failed, in the situation where the patient has relapsed, and/or when repeated exposure to alkylating agents was being considered. This strategy can be used in the management of patients with membranous nephropathy, lupus nephritis, or FSGS.

Monitoring the other potential adverse effects of cyclophosphamide by frequent blood counts, adjusting the dose relative to degree of renal impairment, age, and other comorbid conditions are additional tools to minimize cyclophosphamide toxicity. The likelihood of inducing opportunistic infections such as *pneumocystis jiroveci* pneumonia and/or cytomegalovirus can also be reduced by the use of prophylactic antibiotic/antivirals as described earlier in the "Corticosteroids" section. With respect to the potential to cause infertility in young patients, sperm banking and oocyte cryopreservation can be considered. There are limited data

to suggest a gonadotropin-releasing hormone analog (GnRH-a) may provide ovarian protection, decreasing the rates of premature ovarian failure from 30% to 5% in one study.⁹² However, the use of GnRH-a remains controversial. In the cancer literature, use of GnRH-a is more widely described, yet currently not universally recommended because of conflicting study data.⁹³ There is relatively limited experience with the use of ovarian suppression described in the renal literature. This is, in part, because of the urgency required to start therapy, without sufficient time to institute preventative strategies. Risks should be explained to patients in detail, and alternative therapies considered in patients of childbearing age. Finally, given the teratogenicity of cyclophosphamide, contraception counseling is required for sexually active men and women and the risks and benefits of various modes of contraception should be reviewed. Cryopreservation of sperm may be a realistic option for reproductive-age men requiring urgent therapy.

AZATHIOPRINE

Azathioprine is an inhibitor of inosine monophosphate dehydrogenase, a critical enzyme involved in de novo purine synthesis, required for lymphocyte division resulting in depressed levels of both B and T lymphocytes as well as immunoglobulin synthesis. Its metabolite, 6-thio-guanosine-5'-triphosphate (trio-GTP), causes immunosuppression by blockade of guanosine-5'-triphosphatase activation in T lymphocytes specifically by blocking activation of Rac proteins.⁹⁴

MAJOR ADVERSE EFFECTS

GI side effects including nausea and vomiting are common, and remain the primary reason for treatment interruptions.⁹⁵ Liver toxicity with a significant increase in serum transaminase levels has also been described as has pancreatitis.⁹⁶ Dose-related bone marrow suppression primarily affects white blood cells, but can affect all cell lines, and can be severe in patients with low levels of thiopurine methyltransferase (TPMT). This genetic abnormality affects approximately 0.3% of the population wherein the enzyme is lacking, whereas 11% of individuals are heterozygous for a variant low-activity allele with intermediate activity,⁹⁷ causing diminished azathioprine metabolism. Similarly, allopurinol causes drug accumulation and can result in severe myelosuppression.⁹⁸ As with all immunosuppressive agents, bacterial and viral infections do occur, particularly in the setting of leukopenia, and increased rates of malignancies, in particular skin cancers, have been noted.^{99,100}

STRATEGIES FOR REDUCING TOXICITY

There is no consensus among physicians treating glomerular disease with respect to the assessment for TPMT deficiency. Both genetic testing and functional assays are available, but it is common practice to simply slowly dose escalate while surveying for toxicity. Myelosuppression will typically improve with dose reductions. The gout therapies, allopurinol and febuxostat, are contraindicated when using azathioprine (and other cytotoxic drugs such as cyclophosphamide) due the risk of leukopenia. Monitoring for hepatotoxicity and pancreatitis is also essential when prescribing azathioprine.

MYCOPHENOLIC ACID

Mycophenolate acid (MPA – available as mycophenolate mofetil, or the metabolite mycophenolic acid) is a reversible inhibitor of inosine monophosphate dehydrogenase, a critical enzyme involved in *de novo* purine synthesis, required for lymphocyte division.¹⁰¹ Several factors contribute to the lymphocyte-specific effects of MPA on purine metabolism. First, unlike other cells, lymphocytes are uniquely dependent upon *de novo* purine synthesis to generate RNA and DNA because they do not have a salvage pathway for purine generation. Inhibition of this pathway by MPA, therefore, predominantly affects lymphocyte metabolism. MPA also is a highly potent inhibitor of the isoform of inosine monophosphate dehydrogenase that is expressed in activated lymphocytes (the type II isoform), contributing to its specificity.^{102,103} The selectivity of MPA for inhibiting lymphocyte proliferation is the concept that underlies the reduced toxicity of MPA compared with other alkylating agents that affect all dividing cells. In addition to its effects on T and B cells, MPA may affect fibroblast proliferation/activity¹⁰⁴ and endothelial function.¹⁰⁵

MAJOR ADVERSE EFFECTS

The principal adverse effects of MPA relate to GI symptoms including both upper GI irritation with nausea and vomiting and lower tract involvement with diarrhea. This is more common with MPA than with cyclophosphamide.¹⁰⁶ These symptoms tend to occur early in the course of treatment, and can improve over time.

As with all antimetabolites, MPA can cause hematologic complications including leucopenia and anemia. The myelosuppressive effects of MPA contribute to the risk of infection associated with its use, and although some data suggest a lower infection risk than with cyclophosphamide in lupus nephritis, several studies indicate similar risk of serious infection, and serious infections have been reported.^{85,107–109} Furthermore, the transplantation literature suggests an increased risk of viral infections with MPA, particularly in the context of multidrug regimens.¹¹⁰ Whether there is a difference with respect to risks of late-onset malignancy with MPA when used for the treatment of lupus or other variants of GN is too early to determine.

Although there is no fertility impact, MPA has now clearly emerged as a human teratogen with an identifiable pattern of malformations – craniofacial (microtia or anotia, absent auditory canal, cleft palate, hypertelorism) and limb anomalies.¹¹¹ Manufacturer data note a 33% increased miscarriage rate and at least a 22% rate of teratogenicity. As such, women may elect to terminate the pregnancy if conception occurred during exposure to MPA. The possibility of sperm-mediated MPA teratogenicity has recently been an area of controversy.¹¹² In the absence of clear risk of teratogenicity above population rates, risks should be weighed against the consequences of losing control of disease by altering therapy if conception is planned.¹¹²

STRATEGIES FOR REDUCING TOXICITY

Unlike the transplantation context, the dose of MPA can frequently be titrated up over the course of days to weeks to minimize development of GI symptoms. Splitting the dosage

into four times/day versus the standard two doses/day also reduces GI problems. Temporarily reducing the dose also may be tried, especially in the context of acute GI infections. The predominant effect on the bone marrow is leucopenia and is usually corrected by a temporary dose reduction. If the full dose is still not tolerated, the addition of MPA-sparing agents such as a low dose of steroid or a CNI may be considered. Sexually active women initiating MPA should have a negative pregnancy test before initiating the therapy, and those who desire pregnancy should be off MPA at least 6 weeks before conception. Two forms of contraception should be used by women who are taking MPA.

RITUXIMAB AND OCRELIZUMAB

Rituximab is a genetically engineered, chimeric, murine/human monoclonal antibody directed against the CD20 antigen found on the surface of normal and malignant pre-B and mature B cells.¹¹³ The CD20 antigen is not expressed on hematopoietic stem cells, pro-B cells, normal plasma cells, or other normal tissues. Thus it has an impressive safety record when compared with classic cytotoxic agents. The precise mechanism of action of anti-CD20 antibodies in autoimmune disease and in particular GN is unclear. It is known that B cells play an important role as immunoregulatory cells by both antigen presentation and cytokine release. Their elimination could have dampening effects on other immune cells such as T lymphocytes, dendritic cells, and macrophages. *In vitro* studies have demonstrated that the Fc portion of rituximab binds human complement and can lead to cell lysis of the targeted cell through complement-dependent cytotoxicity and it has been demonstrated that rituximab mediates antibody-dependent cellular cytotoxicity. Most recently, rituximab been shown to be effective in the treatment of idiopathic membranous nephropathy and may work at least in part by depletion of the autoantibody to the podocyte-located antigen phospholipase A₂ receptor.¹¹⁴ Further, rituximab may have a direct podocyte-modulating effect via cross-reactivity with sphingomyelin phosphodiesterase acid-like 3b protein and regulation of acid sphingomyelinase essential for the lipid-raft compartmentalization of the podocyte plasma membrane as well as for the organization and signaling of podocytes in general.¹¹⁵ This potential direct effect on podocytes independent of its known effect on selective depletion of the B-cell clone may make it a very effective option to consider for the treatment of idiopathic glomerular diseases.

Ocrelizumab is a genetically engineered fully humanized monoclonal antibody directed against the same CD20 antigen found on the surface of normal B cells, with the same mode of action as rituximab. Because of its fully humanized nature, this anti-CD20 antibody can be infused more rapidly and has less immediate infusion-related reactions than rituximab. In addition, because of the fully humanized construct, formation of autoantibodies directed against the drug is unlikely, thereby enhancing the potential efficacy and safety of repeated treatments.

MAJOR ADVERSE EFFECTS

Acute infusion-related reactions can vary from minor symptoms to severe life-threatening reactions. Minor reactions

occur in up to 10% of exposed individuals and include skin rash, pruritus, flushing, nausea, vomiting, fatigue, headache, flulike symptoms, dizziness, hypertension, and/or runny nose. Anaphylaxis and shock can occur but are fortunately rare (<1%). Other rare side effects that have been seen with the use of rituximab include anemia, cardiac arrhythmias, respiratory failure, and acute kidney injury (occurring in <0.1%).^{116–120} The latter severe reactions have been seen primarily—although not exclusively—with its use in the treatment of hematologic malignancies where the tumor cell burden is high and an acute tumor lysis syndrome can develop. More delayed adverse effects include serum sickness and an increased incidence of infection including reactivation of latent viral infections including hepatitis B and several cases of pneumocystis.¹²¹ There have been several reports of the development of progressive multifocal leukoencephalopathy (PML) in patients treated with this agent. This devastating syndrome is as a result of the activation of latent JC (John Cunningham) polyomavirus, and is associated with progressive neurologic impairment and ultimately death within months of diagnosis. In the majority of reported cases, PML developed when rituximab was used in combination with additional chemotherapeutic agents.¹²² Given its impact on antibody formation, vaccinations that contain live organisms should be avoided during treatment with rituximab.

Formation of human antichimeric antibodies (HACAs) does occur although their clinical significance is unclear. Despite the appearance of HACA in patients treated with rituximab and their theoretical consequences, these sequelae have not been uniformly observed in the small studies in GN.¹²³ The new fully humanized version of the anti-CD20 agent should either ameliorate or eliminate the issue of HACA formation.

STRATEGIES FOR REDUCING TOXICITY

The risk of acute infusion reaction can be mitigated by careful dose infusion escalation, as well as premedication with antihistamines, and corticosteroids. Administration protocols are available¹²⁴ (see Appendix in McGeoch et al.). The precise dose and/or regimen to use in patients with autoimmune disorders is unknown. Although the relationship between peripheral CD20-positive cell depletion and response is poor, it has been suggested that a single dose, in membranous nephropathy, is adequate for B-cell depletion and provides a similar response in proteinuria as multiple doses of the agent.¹²⁵

ECULIZUMAB

Eculizumab is an anti-C5 (complement factor 5) monoclonal antibody designed for use in diseases characterized by functional impairment of endogenous inhibitors of the activation of the alternative complement pathway. The activation of this cascade has classically been implicated in atypical hemolytic uremic syndrome, thus providing a potential rationale for this agent in this disease and others that might involve the complement pathway.^{126,127} Emerging evidence suggests that complement dysregulation may be a critical contributor to many forms of progressive glomerular injury traditionally attributed to classical or lectin-activated complement pathways. Activation of these pathways and their potential

inhibition by eculizumab may have relevance in the context of C3 nephropathy and dense deposit disease, and perhaps even in the idiopathic immunoglobulin-mediated forms of membranoproliferative glomerulonephritis,¹²⁸ systemic lupus erythematosus,¹²⁹ and other variants of GN, as well as in antibody-mediated rejection in transplantation.¹³⁰

MAJOR ADVERSE EFFECTS

Overall, eculizumab is well tolerated, but there is the potential for an increased risk for infection by encapsulated bacteria such as *Neisseria meningitides*.

STRATEGIES FOR REDUCING TOXICITY

As such, vaccination before instituting therapy and careful surveillance are recommended based on its use in patients with paroxysmal nocturnal hemoglobinuria and in atypical hemolytic-uremic syndrome.¹³¹ In addition, prophylactic antibiotics are widely recommended, particularly when there is concern regarding timing and efficacy of immunization. Guidelines are available to guide choice and duration of antibiotic therapy (e.g., National Renal Complement Therapeutics Centre guidelines).¹³² The precise dose and/or regimen to use in patients with glomerular diseases are presently unknown.

ADRENOCORTICOTROPIC HORMONE

Synthetic formulations of adrenocorticotrophic hormone (ACTH) have previously proven effective in patients with nephrotic syndrome, and the use of ACTH for the treatment of elevated cholesterol and proteinuria dates back many decades.^{133,134} In Europe, the synthetic formulation of ACTH, tetracosactide (Synacthen), has proven similarly effective in both uncontrolled series and a small randomized controlled trial wherein it was compared with cytotoxic therapy.^{135–137} To date, there are limited data to suggest that natural ACTH (H.P. Acthar Gel), which is currently the only available ACTH preparation in North America, may also have a beneficial effect in patients with nephrotic syndrome by lowering proteinuria, improving serum albumin, and cholesterol profiles.¹³⁸

The active ingredients of H.P. Acthar Gel are part of the family of structurally related peptides known as melanocortin peptides. Melanocortin peptides, which include ACTH and the α -, β -, and γ -melanocyte-stimulating hormones, are derived from the natural protein proopiomelanocortin that bind to the cell surface G protein-coupled receptors known as melanocortin receptors (MCRs).¹³⁹ To date, five forms of MCRs have been cloned, each with different tissue distributions, affinities, and physiological roles. As such, its potential therapeutic mechanisms are numerous and complex.¹⁴⁰ Potential renoprotective mechanisms include corticosteroid-mediated systemic immunosuppression and antiinflammatory actions subsequent to ACTH-induced steroidogenesis through MCR 2 interaction as well as direct MCR-mediated immunomodulation and antiinflammatory effects (MCR 1, 3, and 5). Correction of dyslipidemia mediated by MCR 1 and 5 on hepatic cells and neurogenic antiinflammatory effects mediated by MCR 3 and 4 expressed in the central nervous system are likely also beneficial to the kidneys. Finally, a direct

MCR-mediated protective effect on kidney cells, particularly the podocytes, has been described. MCRs are expressed in glomerular podocytes and receptor stimulation has been demonstrated to reduce oxidative stress and improve glomerular morphology by diminishing podocyte apoptosis, injury, and loss in the remnant kidney animal model.¹⁴¹

MAJOR ADVERSE EFFECTS

Although initially thought to be associated with less toxicity than GCs, studies now suggest that the frequency of adverse events associated with this regimen in membranous nephropathy has been underestimated. One open label study reported an incidence of adverse events in 95% of subjects treated with synthetic ACTH.¹⁴² In a recent pilot study side effects were noted to be dose dependent.¹³⁸ Reported side effects associated with the use of ACTH include steroid-like effects including a cushingoid appearance, weight gain, as well as worsening of edema or bloating. Potential skin changes include acne, flushing, and bronzing. With respect to potential psychological effects of the treatment, increased irritability, depression, and improved mood have been noted along with transient insomnia, tremulousness, dizziness, muscle aches or pain, headaches, GI symptoms, blurred vision, as well as generalized weakness or fatigue. Glucose intolerance and frank diabetes are also potential rare side effects that tend to improve with the cessation of treatment.

STRATEGIES FOR REDUCING TOXICITY

The precise dose and/or regimen to use in patients with glomerular diseases are presently unknown, limiting ability to mitigate toxicity. Given the toxicity profile described, similar strategies used in mitigating corticosteroid toxicity should be considered.

TREATMENT ALGORITHMS AND CONSIDERATIONS

The most recent treatment recommendations are reviewed in Chapters 31 and 32. Guidelines assist in decision making, and highlight the evidence supporting different regimens. However, guidelines cannot account for all of the variations in patient and disease variability nor individual preferences in avoiding treatment toxicity (e.g., desire to preserve fertility). Therefore application of these guidelines to the individual patient requires careful consideration. In practice, we consider the following questions before making guideline-based treatment recommendations:

1. What is the risk of progression to kidney failure in this individual patient?
 - The critical question is whether the risks of the medication outweigh the risks of progression of the kidney disease. Complementary considerations are the morbidity and mortality associated with dialysis or with transplantation, where many of the same medications will be used to prevent rejection.
 - Proteinuria is one of our main considerations when we decide whether or not there is an advantage to addition of immunomodulatory treatment to conservative therapy for treatment of patients with primary idiopathic GN.

Although the threshold of proteinuria associated with highest risk of disease progression varies according to disease (Fig. 33.1), this measure is a more important contributor to our treatment decisions than histologic diagnosis. For example, membranous nephropathy may be a lesion that is very amenable to immunologic therapy. However, patients with MGN with persistent subnephrotic range proteinuria have an excellent long-term prognosis.⁶ The added advantage of immunotherapy versus conservative therapy alone is not established in this subpopulation. Renal insufficiency at presentation and progression of renal insufficiency during observation should also be considered. Although this may argue for a more aggressive approach to care, side effects of immunosuppressive therapy are often more frequent in patients with impaired clearance and additional susceptibility to infections. Pathologic indices of kidney injury, in particular tubulointerstitial fibrosis, are also informative to address chronicity and prognosis. Indeed, a common clinical error is to put patients with irreversible kidney injury at risk with little chance of benefit.¹⁴³ Advanced tissue injury with impaired clearance may change the risk–benefit ratio of immunotherapy.

2. What are the patient's comorbid conditions?
 - Ideally the choice of therapy should be tailored to the individual patient, considering comorbid conditions that may place patients at risk of medication toxicity. For example, guidelines suggest that a trial of a minimum of 4 weeks of corticosteroid therapy should be first considered as first-line treatment for idiopathic FSGS associated with clinical features of the nephrotic syndrome. However, in a patient with significant obesity, the weight gain and glucose intolerance induced by steroids are important potential toxicities. A CNI may be a better first choice for that individual patient.
 - Important variables to consider when selecting a patient-specific regimen include age, personal or family history of glucose intolerance, obesity, cancer history, and prior cumulative immunosuppression exposure.
3. What are plans for childbearing?
 - Important considerations in patients contemplating future pregnancy include the teratogenicity of drugs, the effects of the medications on fertility, and the optimization of renal health before pregnancy.
 - Recognizing the complexity of these patients as well as the treatment regimens recommended, treatment of patients with GN in a multidisciplinary specialized clinic, similar to predialysis and transplant clinics, may be advocated. The addition of nursing guidance and supervision, dietary counselling, pharmacy support, as well as social supports may ultimately improve both patient safety and the disease experience of this unique population. Although this setting is not feasible in all health-care centers, the benefits of the multidisciplinary specialized clinic approach merit further study.

CONCLUSION

GN remains a leading cause of CKD and kidney failure. New data regarding clinically relevant surrogate endpoints and therapeutic goals will hopefully facilitate the development and study of novel agents to treat patients with GN and prevent loss

of kidney function. The toxicity of immunotherapeutic agents requires careful consideration by clinicians; the risks of these drugs must always be weighed against the potential benefits in each individual patient, considering patient-, disease-, and drug-specific characteristics. The toxicity and complexity of current regimens also make it imperative that the nephrology community pursue development of novel targeted therapies for these important diseases, and demonstrate efficacy of these treatments in rigorous clinical trials.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Which of the following statements regarding cyclophosphamide is true?
- A cumulative dose of 10g can affect fertility in women.
 - Less cumulative cyclophosphamide exposure can be achieved by using intravenous protocols.
 - Cyclophosphamide can be given safely in elderly patients.
 - All of the above

Answer: d

Rationale: The ovarian toxicity threshold for cyclophosphamide is closely linked to patient age. Doses as low as 10g can cause ovarian failure in patients with limited ovarian reserve. In lupus and vasculitis, IV cyclophosphamide protocols will result in lower cumulative cyclophosphamide. In vasculitis, this has been associated with a higher risk of disease relapse. Observational data suggest that in diseases such as vasculitis, cyclophosphamide can be safely given. However, dosing adjustment is required, particularly in patients with co-existing renal insufficiency.

2. Which of the following can be used to mitigate risk of osteonecrosis in patients receiving corticosteroids?
- Bisphosphonates
 - Calcium and vitamin D
 - Denosumab
 - Calcitonin
 - None of the above

Answer: e

Rationale: While dose and duration of steroid exposure can affect the risk of osteonecrosis, this complication can occur as an idiosyncratic reaction unrelated to dose. No intervention has been shown to prevent this complication.

3. Which of the following statements is true regarding malignancy risk associated with cyclophosphamide?
- The risk of malignancy is lower in patients receiving intravenous administration.
 - The risk threshold for malignancy is a cumulative dose of 45 g of cyclophosphamide.
 - Treatment-associated malignancy typically appears within 5 years of medication exposure.
 - At a cumulative dose of 36 g, the population standardized-incidence ratio of leukemia is as high as 59.
 - Chlorambucil is a more readily accessible and tolerable alternative to cyclophosphamide.

Answer: d

Rationale: The risk of malignancy is more closely related to cumulative dose than method of administration of cyclophosphamide, and there is no known absolute threshold dose. Treatment-associated malignancies appear 7 to 18 years following exposure. Chlorambucil is not available in many countries. While the bladder risk differs, the medication is still associated with morbidity and leukemia risk.